

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM-8

14-07-2026

8. Write the python program to implement DFS.

Aim

To write a Python program to implement the Depth-First Search (DFS) algorithm for graph traversal using user input.

Algorithm

1. Start the program.
2. Define the graph using an adjacency list.
3. Select the starting vertex.
4. Create an empty set to store visited vertices.
5. Mark the current vertex as visited and add it to the traversal list.
6. Check all adjacent vertices of the current vertex.
7. If an adjacent vertex is unvisited, recursively visit that vertex.
8. Continue the process until all reachable vertices are visited.
9. Backtrack whenever there are no unvisited adjacent vertices.
10. Display the DFS traversal order.
11. Generate the graph using NetworkX and Matplotlib.
12. Stop the program.

Program

```
import networkx as nx
import matplotlib.pyplot as plt

g = {'A':['B','C'],'B':['D','E'],'C':['F'],
     'D':[],'E':[],'F':[]}

v = set()
order = []
```

```

def dfs(n):
    v.add(n)
    order.append(n)
    for x in g[n]:
        if x not in v:
            dfs(x)

dfs('A')

print("DFS:", " -> ".join(order))

G = nx.Graph()
for n in g:
    for x in g[n]:
        G.add_edge(n,x)

pos = {'A':(0,2),'B':(-1,1),'C':(1,1),
        'D':(-1.5,0),'E':(-.5,0),'F':(1,0)}

nx.draw(G,pos,with_labels=True,node_size=1800,
        node_color="lightgreen")

for i,n in enumerate(order):
    plt.text(pos[n][0],pos[n][1]+.2,str(i+1),
             fontsize=14)

plt.title("Depth First Search (DFS)")
plt.show()

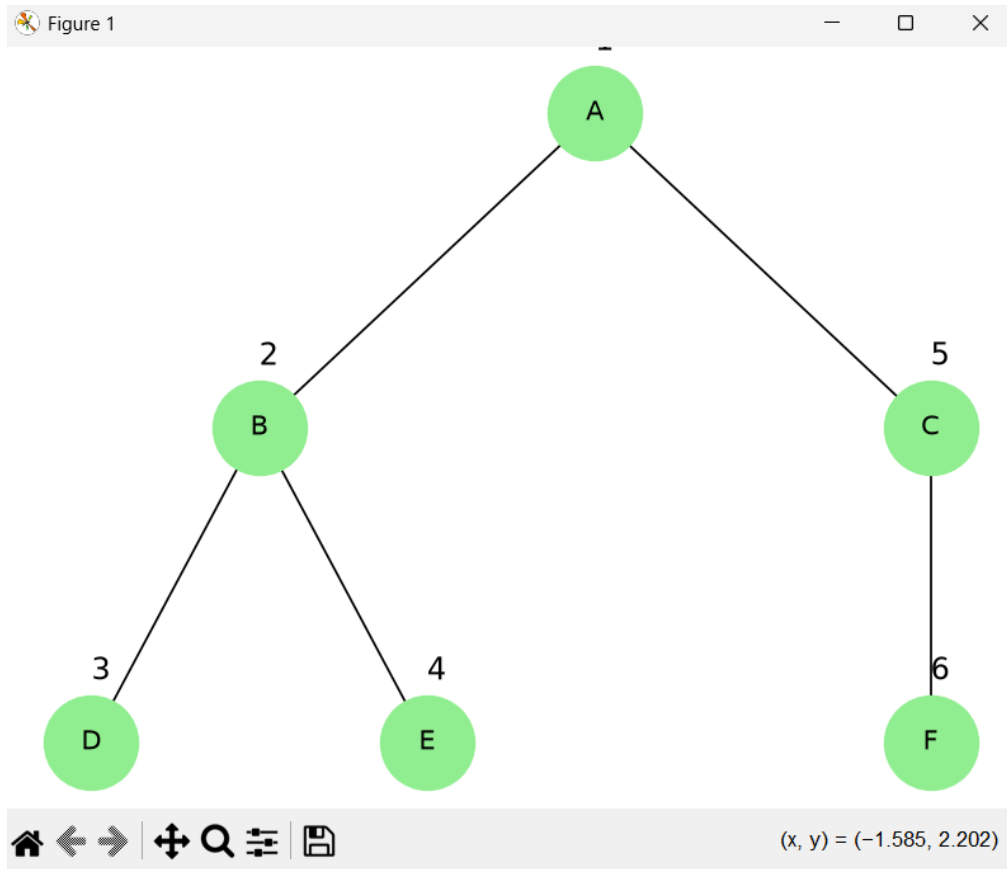
```

Input

Starting node: A

Output

DFS: A -> B -> D -> E -> C -> F



Result:

Thus, the Python program to implement the Depth-First Search (DFS) algorithm using user input was executed successfully, and the graph was traversed in DFS order with the desired output obtained.